THE DERIVED ADJUNCTION RHOM(K, RHOM(L, M)) = RHOM(K TENSOR L, M)

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- For any complexes K, L and I there is a canonical identity of Hom complexes Hom(K, Hom(L, I)) = Hom(Tot(K tensor L), I). invoked at proof line 18702. Provides the termwise adjunction underlying the derived one.

Source reference. Original lemma 0A5Y: sources/stacks-more-algebra/pinned-original-source.tex, source lines 17881–17881. Cited at proof line 18702 as the termwise identity underlying the derived adjunction; the conclusion is used inside the counted span.

Source reference. Original equation 0A63: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18654–18654. Referenced at statement line 18689 as the formula recovered by taking H^0 of the derived adjunction; the formula is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let K, L, M be objects of $D(R)$. There is a canonical isomorphism*

$$R\operatorname{Hom}_R(K, R\operatorname{Hom}_R(L, M)) = R\operatorname{Hom}_R(K \otimes^{\mathbf{L}}_R L, M)$$

in $D(R)$ functorial in K, L, M which recovers (0A63) by taking H^0.

Proof. Choose a K-injective complex $I^\bullet$ representing M and a K-flat complex of R-modules $L^\bullet$ representing L. For any complex of R-modules $K^\bullet$ we have

$$\operatorname{Hom}^\bullet(K^\bullet, \operatorname{Hom}^\bullet(L^\bullet, I^\bullet)) = \operatorname{Hom}^\bullet(\operatorname{Tot}(K^\bullet \otimes_R L^\bullet), I^\bullet)$$

by Lemma 0A5Y. The lemma follows by the definition of $R\operatorname{Hom}$ and because $\operatorname{Tot}(K^\bullet \otimes_R L^\bullet)$ represents the derived tensor product. □

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